

RAVIKULA SILVA

ELECTRICAL GRADUATE ENGINEER - BUILDING SERVICES

Melbourne, VIC · +61 431 638 202 · sravikula@gmail.com · [LinkedIn](#) · [Portfolio](#)

PROFILE SUMMARY

Electrical Building Services Engineer with hands-on experience in the design, estimation, and documentation of electrical building services systems across residential and commercial projects, from concept through to commissioning. Proficient in lighting design (AS 1680, DIALux, DALI, KNX/ETS), electrical load calculations, cable sizing, and protection coordination in compliance with AS 3000, AS 3008, and the NCC. Experienced in producing and updating electrical schematics, SLDs, and shop drawings in AutoCAD, and validating designs using PowerCAD. Proactive and organised team player with a proven ability to coordinate across building services project teams in Australia.

TECHNICAL & CORE COMPETENCIES

Electrical Design & Documentation: AutoCAD, Revit (foundational), PowerCAD, DIALux, Electrical Schematics (SLD) Interpretation, Cable sizing and Protection Coordination

Automation: KNX, ETS, DALI Control, Zencontrol, PLC Programming (Siemens), Python

Standards & Compliance: Australian Standards (AS 3000, AS 3008, AS 1680), NCC

Project Delivery: Project Coordination & Estimation, Site Commissioning, Technical Documentation, Bluebeam Markups

EDUCATION

RMIT University Melbourne, Australia
MSc in Electronic and Electrical Engineering / GPA: 3.6/4 (Distinction) Jul 2025

University of Moratuwa Sri Lanka
BSc (Hons) in Electrical Engineering / GPA: 3.7/4.2 (Second Class Upper Division, equivalent to Distinction) Jun 2022

RELEVANT PROJECTS

Electrical Building Services Design, Residential and Commercial (Point Cook, VIC) Mar 2025 – Jun 2025
RMIT University, Australia

- Designed a complete electrical building services system for a residential dwelling and a detached commercial graphic arts studio, covering HVAC, lighting, and full electrical installation across both project types.
- Sized and verified the submains supply from the residential distribution board to a detached commercial studio 72m away, applying cable sizing and voltage drop calculations to achieve AS 3000 compliance.
- Designed indoor and outdoor lighting systems for residential and commercial spaces using DIALux and AS 1680 illuminance criteria, producing compliant fitting layouts for all functional areas.
- Performed maximum demand calculations, sub-circuit design, cable sizing, protection coordination, and circuit breaker selection to AS 3000 and AS 3008.1 across both residential and commercial installations.
- Validated the full design in PowerCAD achieving an overall voltage drop of 1.25% against a 3.35% permissible limit, confirming compliance across all circuits.
- Produced SLDs and complete electrical documentation including protection device schedules, cable specifications, and circuit layouts for both residential and commercial distribution boards.

WORK EXPERIENCE

iAutomation Pty Ltd Victoria, Australia
Intern Engineer Jan 2026 – Apr 2026

- Supported design and estimation of electrical building services and lighting control systems from concept through to commissioning, contributing to delivery of a commercial healthcare facility valued at AUD 120K.
- Produced and maintained electrical schematics, SLDs, and shop drawings in AutoCAD across active project phases, ensuring documentation accuracy and supporting efficient contractor handover.
- Attended site commissioning activities alongside senior engineers, supporting system verification and sign-off across multiple projects.
- Troubleshoot on-site installation issues alongside senior engineers, identifying and resolving faults to keep project delivery on track.
- Programmed KNX-based control systems using ETS, delivering scene control, DALI dimming, relay switching, and BMS interfacing to meet client lighting automation requirements.
- Coordinated with construction contractors and electrical wholesalers throughout project delivery and commissioning, supporting smooth procurement and on-site execution.

- Tested and validated electrical control systems including VSDs and power supply units using schematics and test equipment, developing hands-on diagnostic skills directly applicable to building services commissioning.
- Assessed semiconductor devices for functional reliability, building a systematic approach to fault identification and component verification applicable to protection and switching systems.

- Automated repetitive engineering workflows using Python scripting, reducing manual effort and improving process consistency across validation tasks.
- Conducted systematic product validation and debugging, producing structured technical documentation that supported clearer identification and resolution of design issues.

- Designed and implemented a capacitor bank system that improved site power factor from 0.85 to 0.95 and reduced reactive power consumption by 12%, demonstrating applied power quality design within a live operational electrical installation.
- Developed PLC-based automated control systems using Siemens S7-300, reducing manual crane operation by 90% and labour costs by 30%, gaining hands-on experience in electrical control design, commissioning, and system integration.

OTHER PROJECTS

- Designed a residential hybrid microgrid integrating a 2.5 kW solar PV array, 2 kW inverter, and 15 kWh battery storage, applying load analysis and system sizing principles applicable to building electrical design.
- Performed techno-economic analysis across four dispatch strategies using HOMER Pro, evaluating NPC, LCOE, and fuel consumption to identify the most cost-effective system configuration.

- Designed and implemented PLC-based automation using Siemens TIA Portal for a modular manufacturing line integrating motor control, pneumatic systems, and sensor-based operations, delivering a fully functional automated system.
- Diagnosed PLC and actuator faults and supported commissioning of control systems, applying structured fault-finding methods to ensure reliable system operation and optimised equipment performance.

PROFESSIONAL QUALIFICATIONS AND CERTIFICATIONS

National Electricity Market (NEM) Foundations AEMO eLearning	Apr 2026
Construction White Card, Australia	Mar 2026
Introduction to ETS6 KNX Association	Sep 2025
Graduate Member, Engineers Australia (GradIEAust)	Since July 2025

LEADERSHIP & VOLUNTEERING

- Directed and delivered the university’s officially recognised career fair across four faculties (Engineering, Architecture, IT, Business), engaging 600+ undergraduates and 80+ companies.
- Led a team of 150+ volunteers across planning, stakeholder coordination, and execution, resulting in 200+ students progressing to next stages in recruitment processes.

- Led media and communications for the final year of a three-year coral reef restoration project, managing content, social media, and stakeholder engagement, while supporting project visibility and impact reporting, contributing to the project receiving a Rotaract Outstanding Project Award (South Asia).

REFERENCES

Jason Wong Head of Lighting Control – iAutomation Pty Ltd +61 458 300 739 jasonw@iautomation.com.au
Don Roshan Project and Systems Engineer – iAutomation Pty Ltd +61 407 133 019 dons@iautomation.com.au